



Tertiary Infotech Academy Pte Ltd

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LESSON PLAN

For

Tableau Desktop Foundations

TGS Ref No: TGS-2025053175

Conducted by

Tertiary Infotech Academy Pte Ltd

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Version 5

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
5	5 September 2026	Retitled to Tableau Desktop Foundations; added Tableau Prep Builder data cleansing to Topic 1 and new Lab 11.	Dr. Alfred Ang

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Course Information

Course Title	Tableau Desktop Foundations
Course Code	TGS-2025053175
TSC	Data Analytics · ATP-PIN-2001-1.1
Duration	2 days · 16 training hours, including 2-hour final assessment
Daily Time	9:00 am–6:00 pm; 1-hour lunch; two 10-minute tea breaks
Mode	Instructor-led concept teaching, demonstrations and Tableau Desktop labs
Trainer	Dr. Alfred Ang

Learning Outcomes

- LO1: Identify and connect appropriate data sources, then prepare trustworthy Tableau data models.
- LO2: Organise fields and create effective visual elements that answer business questions.
- LO3: Extract, combine and analyse data using relationships, joins, unions, filters and calculations.
- LO4: Build, format and share interactive dashboards and workbooks that communicate meaningful insights.

Assessment

- Written Assessment (WA/SAQ) — 4 open-ended questions, 1 hour, open book.
- Practical Performance (PP) — 4 Tableau tasks, 1 hour, open book.
- Assessment runs 4:00–6:00 pm on Day 2.
- Open book: course slides, Learner Guide and approved materials only.
- Learners need at least 75% attendance and a Competent assessment outcome for funding eligibility.

Daily Schedule

Day 1 — Connecting, preparing, exploring and analysing data

Time	Duration	Topic / Lab	Method	Slides
9:00–9:30	30 min	Welcome, attendance, trainer/learner introductions, outcomes and exam orientation	Facilitation	Slides 1–18
9:30–10:30	60 min	Topic 1 — Connecting to and Preparing Data, including Tableau Prep Builder cleansing	Concept + demo	Slides 19–63

10:30–10:40	10 min	Tea break	Break	—
10:40–12:40	120 min	Labs 1–3 — connection/extract audit, relationship model, union and join	Guided hands-on	Slides 55–60
12:40–13:00	20 min	Lab 11 — Tableau Prep cleansing flow: profile, clean, split and Group and Replace	Guided hands-on	Slides 61–62
13:00–14:00	60 min	Lunch break	Lunch	—
14:00–14:55	55 min	Lab 11 continued — pivot, union, join, output the flow and validate against Control	Guided hands-on	Slides 61–62
14:55–15:45	50 min	Topic 2 — Exploring and Analyzing Data	Concept + demo	Slides 64–103
15:45–15:55	10 min	Tea break	Break	—
15:55–17:40	105 min	Labs 4–7 — chart portfolio, organisation/filtering, calculations and geographic analysis	Guided hands-on	Slides 95–102
17:40–18:00	20 min	Day 1 exam check, Q&A and PM digital attendance	Review	Slides 64–103

Total training time excluding lunch: 480 minutes (8 hours), of which 20 minutes are tea breaks counted within the training day and 460 minutes are direct instruction, practice and assessment.

Day 2 — Dashboards, sharing, core concepts and certification readiness

Time	Duration	Topic / Lab	Method	Slides
9:00–9:15	15 min	Day 1 recap and AM digital attendance	Review	Slides 64–103
9:15–10:30	75 min	Topic 3 — Sharing Insights with Dashboards and Workbooks	Concept + demo	Slides 104–132
10:30–10:40	10 min	Tea break	Break	—
10:40–12:30	110 min	Labs 8–9 — interactive dashboard and governed executive sharing	Guided hands-on	Slides 128–131
12:30–13:00	30 min	Topic 4 concepts and certification reasoning	Concept + demo	Slides 133–153
13:00–14:00	60 min	Lunch break	Lunch	—
14:00–15:40	100 min	Lab 10 — Northstar Retail capstone and exam-domain evidence	Guided hands-on	Slides 151–152

15:40–15:50	10 min	Tea break	Break	—
15:50–16:00	10 min	Briefing for Assessment	Assessment	Slides 16
16:00–17:00	60 min	Written Assessment (WA/SAQ) — 4 open-ended questions, open book	Assessment	—
17:00–18:00	60 min	Practical Performance (PP) — 4 Tableau tasks, open book; Assessment attendance	Assessment	—

Total training time excluding lunch: 480 minutes (8 hours), of which 20 minutes are tea breaks counted within the training day and 460 minutes are direct instruction, practice and assessment.

Topic and Lab Slide Map

Topic 01 — Connecting to and Preparing Data in Tableau Desktop • Slides 19–63

Official weighting: 23%. Connections · extracts · metadata · relationships · joins · unions · field properties · Tableau Prep cleansing

Lab 01 — Audit a Retail Sales Connection and Extract • Slides 55–56

Goal: Connect to Excel, validate field metadata, compare live and extract behaviour.

Duration: 60 minutes. Evidence: A validated Tableau data source, field-role audit, live/extract decision and refreshed .hyper extract

Lab 02 — Model Orders, Returns, Products and Targets • Slides 57–58

Goal: Create relationships and compare them with physical joins. Duration: 75 minutes.

Evidence: A four-table logical model with validated cardinality, unmatched-key checks and a join-risk comparison

Lab 03 — Union Monthly Files and Join Store Attributes • Slides 59–60

Goal: Append compatible tables, merge mismatched fields and join a lookup. Duration: 60 minutes. Evidence: A three-month union with provenance fields, merged columns, a validated store join and saved data-source metadata

Lab 11 — Cleanse a Messy Sales Extract with Tableau Prep Builder • Slides 61–62

Goal: Build a repeatable Tableau Prep flow that profiles, cleans, reshapes and outputs an analysis-ready table. Duration: 75 minutes. Evidence: A saved Tableau Prep flow (.tfl) with Input, Clean, Split, Group and Replace, Pivot, Union, Join and Output steps, plus a cleaned .hyper or CSV output and a documented change list

Topic 02 — Exploring and Analyzing Data in Tableau Desktop • Slides 64–103

Official weighting: 37%. Charts · groups · sets · hierarchies · filters · analytics · calculations

Lab 04 — Build the Core Tableau Chart Portfolio • Slides 95–96

Goal: Create and justify nine chart patterns from one realistic sales dataset. Duration: 75 minutes. Evidence: Nine named worksheets: bar, line, scatterplot, symbol map, combined axis, dual axis, stacked bar, density map and highlight table

Lab 05 — Create Groups, Sets, Hierarchies and Filters • Slides 97–98

Goal: Organise customer data and apply filters at the intended analytical stage. Duration: 60 minutes. Evidence: Customer tier groups, Top-N and threshold sets, drill hierarchy, context filter and relative date controls

Lab 06 — Apply Calculations, Table Calculations and Parameters • Slides 99–100

Goal: Create reliable row-level, aggregate and view-dependent analytics. Duration: 75 minutes. Evidence: Calculated KPIs, running totals, percent-of-total, moving average, reference line and parameter-driven measure swap

Lab 07 — Analyse Southeast Asia Logistics Geographically • Slides 101–102

Goal: Create symbol, filled and density maps with trustworthy geographic context. Duration: 60 minutes. Evidence: Resolved geographic roles, three map types, tooltip context and a map-vs-bar decision note

Topic 03 — Sharing Insights with Dashboards and Workbooks • Slides 104–132

Official weighting: 25%. Formatting · dashboards · actions · device layouts · stories · export · publish

Lab 08 — Build an Interactive Service Operations Dashboard • Slides 128–129

Goal: Compose worksheets, containers, filters, tooltips and dashboard actions. Duration: 90 minutes. Evidence: A 1366×768 operational dashboard with KPIs, four views, scoped filters, highlight/filter actions and accessible tooltips

Lab 09 — Optimise, Package and Share an Executive Workbook • Slides 130–131

Goal: Create device layouts, a story, exports and a governed sharing plan. Duration: 60 minutes. Evidence: Desktop and phone layouts, three story points, TWB/TWBX comparison, PDF/PNG/PPT exports and a publishing checklist

Topic 04 — Understanding Tableau Concepts and Certification Preparation • Slides 133–153

Official weighting: 15%. Dimensions · measures · discrete · continuous · aggregation · exam strategy

Lab 10 — Capstone: Northstar Retail Executive Workbook • Slides 151–152

Goal: Integrate all four exam domains and complete certification-readiness evidence. Duration: 100 minutes. Evidence: A validated multi-table model, eight analytical

worksheets, executive dashboard, phone layout, story, packaged workbook and exam-domain checklist

Resources Required

- Tableau Desktop (recommended version matching the certification guide where available)
- Tableau Prep Builder (for the Topic 1 data-cleansing lab)
- Microsoft Excel or compatible viewer
- Eleven individual lab folders with Excel workbooks, starter.twb and acceptance checklists
- Learner Guide and course slides
- LMS/TMS access for courseware, attendance and assessment